BLOCKCHAIN TECHNOLOGY

Clear revision notes | Easy definitions | Smart contracts | Solidity | Formulas | Viva

Prepared from the Blockchain portions of your uploaded ML_BT Assignment Write-up (Assignments 15–19). Topics are reorganized by concept, not in PDF order. Mathematical expressions are written in plain text for readability.

Scope note: The source write-up covers MetaMask and Ether transactions (15–16), a bank smart contract (17), student records using struct/array/fallback and gas (18), and blockchain types/use cases (19). It does not include Assignments 13 and 14.

1. Blockchain basics

What is Blockchain?

Blockchain is a data structure and distributed ledger that stores transaction records in linked blocks. Copies of the ledger are maintained by network participants. It is designed to provide security, transparency, and decentralization.

What is a Distributed Ledger?

A ledger is a record of transactions. A distributed ledger is shared and maintained across multiple computers or participants instead of being kept by only one central authority.

How does a blockchain work? (simple steps)

• A participant creates a transaction, such as sending digital currency.

• The transaction is digitally signed using a private key.

• The transaction is broadcast to the network.

• Network participants/nodes validate transactions according to the network rules.

• A consensus process helps the network agree on accepted transactions and blocks.

• The accepted block is linked to earlier blocks and the shared ledger is updated.

Important terms

Term	Easy definition
Block	A group of transaction data stored together in a blockchain.
Chain	The ordered connection of blocks; each block refers to the previous block using its hash.
Node	A computer or participant that takes part in a blockchain network.
Distributed ledger	A shared record maintained across multiple network participants.
Decentralization	Control is distributed; no single central authority controls the whole network.
Consensus	The process by which network participants agree on valid transactions or blocks.
Immutability	Accepted records are difficult to change; changes can be detected and may require network agreement.
Transparency	Transactions or records can be inspected by participants according to the network's access rules.
Tamper evidence	Unauthorized changes can be detected, for example through hashes and block links.

2. Main features, benefits and limitations

Feature	Meaning
Decentralized	No single person or group has full control of the shared network.
Immutable / hard to alter	Once accepted, historical records are difficult to change.
Tamper-evident	Changing data can change hashes and break the expected block links.
Digital signatures	Help prove that a transaction was authorized by the holder of the relevant private key.
Consensus	Helps participants agree on which transactions or blocks are accepted.
Shared history	Participants can refer to a common transaction history.

Benefits mentioned in the write-up

• Time-saving: reducing the need for a central authority or intermediary to verify every settlement.

• Cost-saving: fewer intermediaries and shared records can reduce some transaction and verification costs.

• Security: cryptographic techniques and distributed records make unauthorized changes easier to detect.

Limitations / practical considerations

• Public networks may process transactions more slowly when many nodes must participate in verification.

• Proof of Work can consume significant energy and computing resources.

• Private and consortium networks require participants to trust the organizations that govern access and operation.

• Blockchain is not automatically immune to fraud, stolen keys, software bugs, or poor smart-contract design.

The source describes blockchain as tamper-proof; for exam understanding, say records are designed to be tamper-evident and very difficult to alter, not that every blockchain is impossible to attack.

3. Types of blockchain

Type	Who can participate?	Advantages / uses in the source
Public	Open to anyone; generally permissionless.	Decentralized, broad participation, shared verification. Examples: Bitcoin and Ethereum.
Private	Only authorized users; controlled by one organization.	Faster verification, configurable access and privacy. Uses: internal auditing and asset management. Examples mentioned: Hyperledger, Corda.
Hybrid	Combines private control with selected public visibility.	Can keep some data private while making selected information publicly verifiable. Uses mentioned: healthcare, government, real estate and finance.
Consortium (Federated)	Governed by multiple organizations.	Shared control among organizations; useful for banking, payment processing and supply-chain collaboration. Examples mentioned: Tendermint and Multichain.

Quick comparison

Point	Public	Private	Hybrid	Consortium
Access	Anyone	Authorized users	Mixed access	Selected member organizations
Control	Distributed network	One organization	Organization plus public-facing elements	Multiple organizations
Typical focus	Open participation	Privacy and internal control	Selective transparency	Inter-organization collaboration

4. Cryptography, hashes and Merkle Trees

Cryptographic hash

A hash function takes input data of any size and produces a fixed-length output called a hash or digest. A small change in input normally produces a very different hash. Blockchain uses hashes to help link blocks and detect data changes.

```
Input data --hash function--> fixed-length hash
```

Previous block hash

A block stores a reference to the previous block's hash. If earlier block data is changed, its hash changes and the following block's reference no longer matches the expected value.

Digital signature

A digital signature is created using a private key and can be checked using the corresponding public key. It helps verify that a transaction was authorized by the holder of that private key.

Public key and private key

Key	Meaning
Public key / address	Can be shared so others can identify an account or verify signatures. An account address is derived from public-key information.
Private key	Secret key used to authorize/sign transactions. It must never be shared.
Seed phrase / recovery phrase	A sequence of words used to restore access to a wallet. Anyone who gets it may be able to control the wallet.

Merkle Tree

A Merkle Tree is a tree structure made from hashes. Transaction hashes are paired and hashed repeatedly until one final hash, called the Merkle Root, remains. The Merkle Root summarizes the transactions and helps verify data integrity efficiently.

```
Transaction hashes -> pair and hash -> higher-level hashes -> Merkle Root
```

Double spending

Double spending means attempting to spend the same digital currency value more than once. Transaction validation and the network's consensus mechanism help prevent conflicting spends from both being accepted.

5. Cryptocurrency and Ether

Cryptocurrency

A cryptocurrency is a digital asset/payment system that uses cryptography to secure transactions. Transactions are recorded on a distributed ledger, and users control assets through cryptographic wallet keys.

How cryptocurrency works

• Transactions are recorded on a blockchain or another distributed ledger.

• Users hold keys in a digital wallet to authorize transactions.

• Transactions are broadcast to the network and validated according to its rules.

• Some cryptocurrencies are created through mining; users may also acquire them through other methods.

Name	Short description from the source
Bitcoin (BTC)	Launched in 2009; the first and best-known cryptocurrency. Its main focus is peer-to-peer digital money.
Ethereum	A blockchain platform launched in 2015 that supports smart contracts and decentralized applications.
Ether (ETH)	The native cryptocurrency of the Ethereum network.
Litecoin	A cryptocurrency similar to Bitcoin, described in the source as aiming for faster payments and more transactions.
Ripple / XRP	The source describes Ripple as a distributed ledger system used to track different kinds of transactions, including work with financial institutions.
Altcoins	A general term for cryptocurrencies other than Bitcoin.

Bitcoin vs Ethereum

Bitcoin	Ethereum
Primarily designed for peer-to-peer digital money.	A programmable blockchain platform that supports smart contracts and decentralized applications.
Native asset: BTC.	Native asset: ETH (Ether).

Cryptojacking

Cryptojacking is the unauthorized use of another person's device or computing resources to mine cryptocurrency, often without the device owner's knowledge.

Four common cryptocurrency scams from the revision material

Scam	Meaning
Ponzi / investment scam	Old investors are paid using funds from new investors while fake profits are promised.
Phishing	Fake websites, messages or emails try to steal passwords, private keys or recovery phrases.
Fake ICO / token	A fraudulent project collects money using false claims or promises.
Pump-and-dump	Promoters artificially raise a coin's price and then sell their holdings, leaving later buyers with losses.

6. Crypto wallets and MetaMask (Assignments 15–16)

What is a crypto wallet?

A crypto wallet is software or a physical device that manages the keys used to access and authorize transactions involving cryptocurrency. The assets themselves are recorded on the blockchain; the wallet manages the keys.

Wallet type	Meaning
Hot wallet	Connected to the internet; convenient for frequent use but exposed to online risks.
Cold wallet	Keeps keys offline, often using a hardware device; reduces exposure to online attacks.

What is MetaMask?

MetaMask is a browser wallet that lets users manage Ethereum accounts and interact with Ethereum-based applications. It can display assets and transaction activity, send transactions, and connect to decentralized applications (dApps).

Basic MetaMask procedure from the practical

• Install the MetaMask browser extension from the official browser extension store.

• Open the extension and create a wallet/account with a password.

• Reveal and securely back up the Secret Recovery Phrase; verify it when prompted.

• Open the account view and copy the public wallet address when you need to receive funds.

• Select the correct blockchain network before sending or receiving assets.

• To send crypto, choose Send, enter the recipient address and amount, review the network fee, then confirm or reject the transaction.

• Use the Assets tab to view assets and Activity to view transaction history.

• To use a dApp, choose Connect Wallet and review the connection request before approving.

Wallet safety: Never share your Secret Recovery Phrase or private key. Verify the recipient address, network and transaction details before confirming. A wallet password is not a substitute for protecting the recovery phrase.

Advantages of MetaMask listed in the source

• Popular: commonly supported by many dApps.

• Simple: helps users manage accounts and sign transactions through the wallet interface.

• Saves space: users do not need to download the entire Ethereum blockchain to their computer.

• Integrated: many dApps provide a Connect Wallet option for interacting with MetaMask.

E-transfer safety

The source notes that electronic transfers are generally safer when using trusted banking applications, encryption, OTP/PIN protection and verified recipient details. Never share OTPs or UPI PINs, avoid unknown links and report suspicious transactions promptly.

7. Smart contracts and Solidity

Smart contract

A smart contract is a program stored on a blockchain. Its functions can be executed through blockchain transactions or calls. In the practicals, smart contracts are used to manage bank deposits/withdrawals and student records.

Solidity

Solidity is a high-level, statically typed programming language used to write smart contracts for Ethereum and other EVM-compatible blockchain networks.

Important Solidity terms

Term	Easy definition
contract	A Solidity program containing data and functions.
uint / uint256	Unsigned integer types; they store non-negative whole numbers.
address	A data type used to store an account or contract address.
string	A text value.
mapping	A key-value data structure; example: mapping(address => uint256) balanceOf.
struct	A user-defined type that groups related fields, possibly of different data types.
array	A collection of values of the same declared type; a dynamic array can grow.
public	Makes a state variable or function accessible through a generated/external interface, subject to Solidity rules.
payable	Allows a function or address to receive Ether.
msg.sender	The address that called the current function.
msg.value	The amount of Ether sent with the current payable call, in wei.
require()	Checks a condition; if it is false, the transaction reverts.
view	Marks a function that reads state but does not modify it.
fallback()	Special function called when calldata does not match another function, or when Ether is sent and no suitable receive function handles it.

The examples in the original practical use an older Solidity version. Syntax can differ in current compiler versions; follow the version specified by your lab setup when compiling.

8. Bank account smart contract (Assignment 17)

Objective

Create and deploy a smart contract on an Ethereum test network to deposit money, withdraw money and show the account balance.

How the contract works

- A mapping stores each account address and its assigned balance.
- The deposit function is payable so it can receive Ether.
- The contract checks that the amount supplied matches the Ether sent (msg.value).
- The balance for msg.sender is increased after a valid deposit.
- The withdraw function checks that the caller has enough balance.
- The stored balance is reduced before Ether is transferred to the caller.
- A balance lookup displays the amount assigned to the caller/account.

```
mapping(address => uint256) public balanceOf;
require(msg.value == amount);
balanceOf[msg.sender] += amount;
require(amount <= balanceOf[msg.sender]);
balanceOf[msg.sender] -= amount;
```

Why update balance before transferring Ether?

To reduce reentrancy risk. If an external contract receives Ether, its code may run and call back into the original contract. Updating the balance first helps prevent the same balance from being withdrawn repeatedly.

Checks-Effects-Interactions pattern

Stage	Meaning in withdraw()
Checks	Check that the requested amount is not greater than the caller's balance.
Effects	Reduce the caller's stored balance.
Interactions	Transfer Ether to the caller after updating the state.

Reentrancy

Reentrancy is a smart-contract vulnerability in which an external contract calls back into the original contract before the original operation has safely completed. The Checks-Effects-Interactions pattern helps reduce this risk.

Mapping

A mapping stores key-value pairs. In mapping(address => uint256), an address is the key and an unsigned integer is the value. A mapping value that has not been set is zero by default. The mapping tracks assigned balances; it does not itself transfer Ether.

9. Student records smart contract (Assignment 18)

Objective

Create and deploy a Solidity contract that stores student records using a struct and array, includes a payable fallback function, and observes gas and transaction fees.

Struct example

```
struct Student {
uint id;
string name;
uint age;
string course;
}
```

A Student struct groups the student's ID, name, age and course into one record.

Array example

```
Student[] public students;
students.push(Student(id, name, age, course));
```

Student[] is a dynamic array that can store multiple Student records. push() adds a new record to the array.

Fallback function example

```
fallback() external payable { }
```

The fallback function is a special function that runs when a call does not match an existing function, or when Ether is sent and no suitable receive function handles it. The payable keyword allows it to receive Ether.

Practical procedure

• Open Remix IDE and create StudentData.sol.

• Write the Student struct, dynamic Student array, add/retrieve functions and payable fallback function.

• Compile using the appropriate Solidity compiler version.

• Open Deploy & Run Transactions in Remix.

• Choose the required environment, such as an injected wallet connected to an Ethereum test network.

• Deploy the contract and approve the transaction in the wallet.

• Observe deployment details such as gas used, gas limit, effective gas price, transaction hash and contract address.

• Call addStudent() to insert records and getStudent() or the public array getter to retrieve them.

• Record the gas consumed and transaction fee; compare deployment with a student insertion.

10. Gas and transaction fee

What is gas?

Gas is the unit used by Ethereum to measure the computational work required to execute operations.

Term	Meaning
Gas used	The amount of computational gas consumed by the transaction.
Gas limit	The maximum gas the sender allows the transaction to consume.
Gas price	The price paid per unit of gas; the effective price is used in the transaction fee calculation.
Transaction fee	The fee paid for the computation and processing of the transaction.
Transaction hash	A unique identifier for a blockchain transaction.
Contract address	The address where a successfully deployed smart contract can be accessed.

```
Transaction Fee = Gas Used * Effective Gas Price
```

Example from the source:

```
Gas used = 100,000
Gas price = 20 Gwei
Fee = 100,000 * 20 Gwei = 2,000,000 Gwei = 0.002 ETH
```

Actual gas values and fees vary with the network and current conditions. Contract deployment often costs more than a simple data insertion because deployment stores the contract bytecode and initializes contract state.

11. Tools used in the blockchain practicals

Tool / technology	Purpose
MetaMask	Browser wallet used to manage accounts, sign transactions, view activity and connect to dApps.
Remix IDE	Browser-based development environment used in the practical to write, compile, deploy and test Solidity contracts.
Solidity	Programming language used to write Ethereum smart contracts.
Ethereum test network	A test blockchain environment used to deploy and test contracts without using real mainnet funds.
Ether (ETH)	Ethereum's native asset; used in transactions and to pay network fees.
EVM	Ethereum Virtual Machine; the execution environment for Ethereum smart contracts.

12. Assignment-wise quick revision

Assignment	Practical / title	Key concepts to revise
15	Install MetaMask and study Ether spending per transaction	Blockchain basics, features, benefits, MetaMask setup, sending Ether, Merkle Tree, double spending, blockchain applications.
16	Create a MetaMask wallet for crypto transactions	Cryptocurrency, how it works, Bitcoin, Ethereum, Litecoin, Ripple, hot/cold wallets, scams, cryptojacking, e-transfer safety.
17	Bank account smart contract	Solidity, mapping, deposit, withdraw, balance, msg.sender, msg.value, require, payable, reentrancy.
18	Student data using struct, array and fallback	Struct, dynamic array, fallback, payable, Remix, deployment, gas, transaction fee.
19	Survey of blockchain types and real-time use cases	Public, private, hybrid and consortium blockchains; advantages, disadvantages and applications.

13. Short viva questions and answers

1. What is blockchain? A distributed ledger that stores records in linked blocks across network participants.

2. What is a distributed ledger? A shared record of transactions maintained by multiple participants.

3. What is decentralization? Control is distributed instead of being held by one central authority.

4. What is consensus? A process through which network participants agree on accepted transactions or blocks.

5. What is immutability? Accepted records are difficult to change and changes can be detected.

6. What is a hash? A fixed-length output generated from input data by a hash function.

7. What is a digital signature? A cryptographic proof that helps verify a transaction was authorized by a key holder.

8. What is a Merkle Tree? A tree of hashes that combines transaction hashes into one Merkle Root.

9. What is double spending? Trying to spend the same digital currency value more than once.

10. What is cryptocurrency? A digital asset/payment system that uses cryptography to secure transactions.

11. What is Bitcoin? The first and best-known cryptocurrency, launched in 2009.

12. What is Ethereum? A programmable blockchain platform that supports smart contracts and dApps.

13. What is Ether? The native cryptocurrency of the Ethereum network.

14. What is MetaMask? A browser wallet used to manage Ethereum accounts and connect to dApps.

15. What is a hot wallet? A wallet connected to the internet.

16. What is a cold wallet? A wallet that keeps keys offline, often using a hardware device.

17. What is cryptojacking? Unauthorized use of someone else's computing resources to mine cryptocurrency.

18. What is a smart contract? A program stored on a blockchain and executed through transactions or calls.

19. What is Solidity? A high-level, statically typed language for writing Ethereum smart contracts.

20. What is mapping? A key-value data structure, such as an address mapped to a balance.

21. What is struct? A user-defined type that groups related fields of different data types.

22. What is an array? A collection of values of the same declared type.

23. What is payable? A Solidity keyword that allows a function or address to receive Ether.

24. What is msg.sender? The address that called the current function.

25. What is msg.value? The Ether amount sent with the current payable call, measured in wei.

26. What is require()? A function that checks a condition and reverts if it is false.

27. What is a fallback function? A special function called when no function matches the call or no suitable receive function handles incoming Ether.

28. What is gas? A unit measuring computational work on Ethereum.

29. What is gas limit? The maximum gas the sender allows a transaction to consume.

30. What is reentrancy? A vulnerability where an external contract calls back into the original contract before it safely completes an operation.

31. What is the Checks-Effects-Interactions pattern? Check conditions, update contract state, then make external interactions.

32. What is a public blockchain? A blockchain open to anyone who wants to participate.

33. What is a private blockchain? A blockchain where participation is limited to authorized users.

34. What is a hybrid blockchain? A combination of private control and selected public visibility.

35. What is a consortium blockchain? A blockchain governed by multiple organizations.

14. Important formulas and final reminders

Topic	Formula / point
Transaction fee	Gas Used * Effective Gas Price
Wei and Gwei	1 Gwei = 1,000,000,000 wei; 1 ETH = 1,000,000,000 Gwei
Merkle Tree	Pair transaction hashes, hash each pair, repeat until one Merkle Root remains.
Wallet transaction	Sender signs transaction -> network validates -> transaction is included/recorded.
Bank deposit	Check msg.value == amount; then add amount to balanceOf[msg.sender].
Bank withdrawal	Check amount <= balance; reduce balance first; then transfer Ether.

• Never share a private key or Secret Recovery Phrase.

• Always verify the recipient address and blockchain network before sending cryptocurrency.

• Remember: wallet keys control access; the blockchain records the assets and transactions.

• For a bank contract, know mapping, deposit, withdraw, require, payable and reentrancy.

• For a student contract, know struct, array, fallback, gas used, gas limit and transaction fee.

• For Assignment 19, learn all four blockchain types with their access rules and use cases.

Source note: The assignment mapping, terminology, examples and practical steps are based on the Blockchain sections (Assignments 15–19, pages 28–42) of your uploaded ML_BT Assignment Write-up. Explanations have been simplified and reorganized for revision. The source's Solidity examples use an older compiler version, so exact syntax may differ in a current Remix compiler.